$$\lambda_6 \to \frac{1}{16\pi^2} (\frac{2}{9} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{3,0}[m_2] + \frac{1}{3} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{4,-1}[m_2] -$$
$$\frac{16}{45} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{5,-2}[m_2] + \frac{1}{36} \overline{C_{\phi2\phi1}} g_2^2 (-9 \overline{y_e}^{pr} y_e^{pr} + 4 \sum_p g_2^2) \mathrm{LF}_{3,0}[m_{\tilde{l}}^p] +$$
$$\frac{1}{24} \overline{C_{\phi2\phi1}} g_2^2 (6 \overline{y_e}^{pr} y_e^{pr} - 5 \sum_p g_2^2) \mathrm{LF}_{4,-1}[m_{\tilde{l}}^p] + \frac{4}{45} \sum_p \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{5,-2}[m_{\tilde{l}}^p] +$$
$$\frac{1}{12} \overline{C_{\phi2\phi1}} g_2^2 (-9 \overline{y_d}^{pr} y_d^{pr} + 9 \overline{y_u}^{pr} y_u^{pr} + 4 \sum_p g_2^2) \mathrm{LF}_{3,0}[m_{\tilde{q}}^p] -$$
$$\frac{1}{8} \overline{C_{\phi2\phi1}} g_2^2 (-6 \overline{y_d}^{pr} y_d^{pr} + 6 \overline{y_u}^{pr} y_u^{pr} + 5 \sum_p g_2^2) \mathrm{LF}_{4,-1}[m_{\tilde{q}}^p] + \frac{4}{15} \sum_p \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{5,-2}[m_{\tilde{q}}^p] +$$
$$\frac{1}{9} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{3,0}[\tilde{\mu}] + \frac{1}{6} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{4,-1}[\tilde{\mu}] - \frac{8}{45} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{5,-2}[\tilde{\mu}] - m_1 \tilde{\mu} g_1^4 \mathrm{LF}_{2,2,-1}[m_1, \tilde{\mu}] -$$
$$\frac{4}{3} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{2,1,0}[m_2, \tilde{\mu}] - \frac{1}{12} g_2^4 (13 \overline{C_{\phi2\phi1}} + 36 m_2 \tilde{\mu}) \mathrm{LF}_{2,2,-1}[m_2, \tilde{\mu}] +$$
$$\frac{1}{6} m_2 g_2^4 (-3 m_2 \overline{C_{\phi2\phi1}} + 2 \tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2})) \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] +$$
$$\frac{2}{3} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \frac{2}{3} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{3,2,-2}[m_2, \tilde{\mu}] -$$
$$\frac{1}{6} m_2 \tilde{\mu} g_2^4 (3 C_{\phi1^2} + C_{\phi2^2}) \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{d}}^r, m_{\tilde{q}}^p] -$$
$$\frac{1}{4} g_1^2 (\overline{C_{\phi2\phi1}} \overline{a_d}^{pr} a_d^{pr} + \tilde{\mu} \overline{y_d}^{pr} (a_d^{pr} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr})) \mathrm{LF}_{2,2,0}[m_{\tilde{d}}^r, m_{\tilde{q}}^p] -$$
$$\frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{e}}^r, m_{\tilde{l}}^p] -$$
$$\frac{1}{4} g_1^2 (\overline{C_{\phi2\phi1}} \overline{a_e}^{pr} a_e^{pr} + \tilde{\mu} \overline{y_e}^{pr} (a_e^{pr} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr})) \mathrm{LF}_{2,2,0}[m_{\tilde{e}}^r, m_{\tilde{l}}^p] +$$
$$\frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e}^{pr} a_e^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] - \frac{1}{4} \tilde{\mu} \overline{y_e}^{pr} a_e^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] +$$
$$\frac{1}{4} (\overline{C_{\phi2\phi1}} \overline{a_e}^{pr} a_e^{pr} (g_1^2 - g_2^2) + 2 \tilde{\mu} g_1^2 \overline{y_e}^{pr} (a_e^{pr} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr})) \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] +$$
$$\frac{1}{4} g_1^2 (\overline{C_{\phi2\phi1}} \overline{a_e}^{pr} a_e^{pr} + \tilde{\mu} \overline{y_e}^{pr} (a_e^{pr} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr})) \mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] +$$
$$\frac{1}{8} (3 \overline{C_{\phi2\phi1}} \overline{a_e}^{pr} a_e^{pr} (-g_1^2 + g_2^2) -$$
$$\tilde{\mu} \overline{y_e}^{pr} (3 a_e^{pr} (C_{\phi1^2} + C_{\phi2^2}) (3 g_1^2 + g_2^2) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr} (9 g_1^2 + g_2^2))) \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] +$$
$$\frac{1}{6} (-\overline{C_{\phi2\phi1}} g_2^2 \overline{a_e}^{pr} a_e^{pr} + \tilde{\mu} \overline{y_e}^{pr} (a_e^{pr} (3 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) + 3 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_e^{pr}))$$
$$\mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] + \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_d}^{pr} a_d^{pr} \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] -$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{pr} a_d^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] +$$
$$\frac{1}{4} (-\overline{C_{\phi2\phi1}} \overline{a_d}^{pr} a_d^{pr} (g_1^2 + 3 g_2^2) + 2 \tilde{\mu} g_1^2 \overline{y_d}^{pr} (a_d^{pr} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr}))$$
$$\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] + \frac{1}{4} g_1^2 (\overline{C_{\phi2\phi1}} \overline{a_d}^{pr} a_d^{pr} + \tilde{\mu} \overline{y_d}^{pr} (a_d^{pr} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr}))$$
$$\mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] + \frac{3}{8} (\overline{C_{\phi2\phi1}} \overline{a_d}^{pr} a_d^{pr} (g_1^2 + 3 g_2^2) -$$
$$\tilde{\mu} \overline{y_d}^{pr} (a_d^{pr} (C_{\phi1^2} + C_{\phi2^2}) (5 g_1^2 + 3 g_2^2) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr} (5 g_1^2 + g_2^2))) \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] +$$
$$\frac{1}{2} (-\overline{C_{\phi2\phi1}} g_2^2 \overline{a_d}^{pr} a_d^{pr} + \tilde{\mu} \overline{y_d}^{pr} (a_d^{pr} (3 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) + 3 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_d^{pr}))$$
$$\mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] + \frac{3}{4} \overline{C_{\phi2\phi1}} \overline{y_u}^{st} y_u^{pt} (\overline{y_d}^{pr} y_d^{sr} - 2 \overline{y_u}^{pr} y_u^{sr}) \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s] -$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \overline{y_u}^{st} y_u^{pt} (\overline{y_d}^{pr} y_d^{sr} - 2 \overline{y_u}^{pr} y_u^{sr}) \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s] -$$
$$\frac{1}{4} \tilde{\mu} \overline{y_u}^{pr} a_u^{pr} (g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^p, m_{\tilde{u}}^r] + \frac{1}{8} (-\overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{pr} (g_1^2 - 9 g_2^2) + \tilde{\mu} \overline{y_u}^{pr}$$
$$(a_u^{pr} (-g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + g_2^2 (9 C_{\phi1^2} + 5 C_{\phi2^2})) - \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr} (g_1^2 - 5 g_2^2))) \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^p,$$
$$m_{\tilde{u}}^r] + \frac{3}{4} g_2^2 (\overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{pr} + \tilde{\mu} \overline{y_u}^{pr} (a_u^{pr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr})) \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^p, m_{\tilde{u}}^r] -$$
$$\frac{1}{4} g_2^2 (3 \overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{pr} + 2 \tilde{\mu} \overline{y_u}^{pr} (a_u^{pr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr})) \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^r] -$$
$$\frac{3}{4} g_2^2 (\overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{pr} + \tilde{\mu} \overline{y_u}^{pr} (a_u^{pr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr})) \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^r] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{2,1,0}[m_{\tilde{q}}^s, m_{\tilde{q}}^p] - \frac{3}{4} \overline{C_{\phi2\phi1}} \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} \mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p] +$$
$$\frac{1}{4} \tilde{\mu} \overline{y_u}^{pr} a_u^{pr} (7 g_1^2 + 3 g_2^2) \mathrm{LF}_{2,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] - \frac{3}{4} \tilde{\mu} \overline{y_u}^{pr} a_u^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \frac{1}{4}$$
$$(7 \overline{C_{\phi2\phi1}} g_1^2 \overline{a_u}^{pr} a_u^{pr} + \tilde{\mu} \overline{y_u}^{pr} (a_u^{pr} (7 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr} (4 g_1^2 - g_2^2)))$$
$$\mathrm{LF}_{3,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \frac{1}{8} (\overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{pr} (g_1^2 - 9 g_2^2) +$$
$$\tilde{\mu} \overline{y_u}^{pr} (a_u^{pr} (g_1^2 (C_{\phi1^2} + C_{\phi2^2}) - g_2^2 (9 C_{\phi1^2} + 5 C_{\phi2^2})) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr} (g_1^2 - 5 g_2^2)))$$
$$\mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \frac{1}{4} (-15 \overline{C_{\phi2\phi1}} g_1^2 \overline{a_u}^{pr} a_u^{pr} + 3 \tilde{\mu} \overline{y_u}^{pr}$$
$$(-a_u^{pr} (5 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pr} (-2 g_1^2 + g_2^2))) \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] +$$
$$\frac{1}{2} (3 \overline{C_{\phi2\phi1}} g_1^2 \overline{a_u}^{pr} a_u^{pr} + \tilde{\mu} \overline{y_u}^{pr} (a_u^{pr} (3 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) - \overline{C_{\phi2\phi1}} \tilde{\mu} g_2^2 y_u^{pr}))$$
$$\mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 (g_1^2 + g_2^2) \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_1] -$$
$$\frac{1}{4} g_1^2 (g_1^2 + g_2^2) (2 \overline{C_{\phi2\phi1}} + m_1 \tilde{\mu}) \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_1] +$$
$$\frac{1}{4} m_1 \tilde{\mu} g_1^2 (C_{\phi1^2} + C_{\phi2^2}) (g_1^2 + g_2^2) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] + \frac{3}{4} \overline{C_{\phi2\phi1}} g_1^4 \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_1] +$$
$$\frac{1}{4} g_1^4 (-2 m_1 \tilde{\mu} (4 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} (3 m_1^2 + 2 \tilde{\mu}^2)) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] +$$
$$\frac{1}{2} C_{\phi2\phi1} g_1^4 m_1^2 \tilde{\mu}^2 \mathrm{LF}_{3,2,0}[\tilde{\mu}, m_1] + \frac{1}{6} \overline{C_{\phi2\phi1}} g_1^2 (6 g_1^2 + g_2^2) \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1] -$$
$$\frac{1}{8} m_1 \tilde{\mu} g_1^2 (6 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + g_2^2 (3 C_{\phi1^2} + 5 C_{\phi2^2})) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] -$$
$$\frac{1}{2} \overline{C_{\phi2\phi1}} g_1^4 \mathrm{LF}_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 (m_1 \tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2}) - \overline{C_{\phi2\phi1}} (m_1^2 + \tilde{\mu}^2)) \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] -$$
$$\frac{1}{2} C_{\phi2\phi1} g_1^4 m_1^2 \tilde{\mu}^2 \mathrm{LF}_{4,2,-1}[\tilde{\mu}, m_1] + \frac{1}{6} \overline{C_{\phi2\phi1}} g_1^2 (-3 g_1^2 + g_2^2) \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1] +$$
$$\frac{1}{6} m_1 \tilde{\mu} g_1^2 (3 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \frac{3}{4} m_2 \tilde{\mu} g_2^2 (g_1^2 + g_2^2) \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_2] +$$
$$\frac{1}{12} g_2^2 (2 \overline{C_{\phi2\phi1}} (-9 g_1^2 + 7 g_2^2) - 9 m_2 \tilde{\mu} (g_1^2 + g_2^2)) \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2] +$$
$$\frac{1}{12} m_2 \tilde{\mu} g_2^2 (9 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + g_2^2 (-3 C_{\phi1^2} + 5 C_{\phi2^2})) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] +$$
$$\frac{49}{12} \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{3,2,-2}[\tilde{\mu}, m_2] +$$
$$\frac{1}{12} g_2^4 (-2 m_2 \tilde{\mu} (45 C_{\phi1^2} + 14 C_{\phi2^2}) + 3 \overline{C_{\phi2\phi1}} (11 m_2^2 + 6 \tilde{\mu}^2)) \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] +$$
$$\frac{3}{2} C_{\phi2\phi1} g_2^4 m_2^2 \tilde{\mu}^2 \mathrm{LF}_{3,2,0}[\tilde{\mu}, m_2] - \frac{3}{2} \overline{C_{\phi2\phi1}} g_2^2 (-2 g_1^2 + g_2^2) \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2] +$$
$$\frac{1}{8} m_2 \tilde{\mu} g_2^2 (-18 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + g_2^2 (3 C_{\phi1^2} - 11 C_{\phi2^2})) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] -$$
$$2 \overline{C_{\phi2\phi1}} g_2^4 \mathrm{LF}_{4,2,-3}[\tilde{\mu}, m_2] - 2 g_2^4 (-m_2 \tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} (m_2^2 + \tilde{\mu}^2)) \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] -$$
$$2 C_{\phi2\phi1} g_2^4 m_2^2 \tilde{\mu}^2 \mathrm{LF}_{4,2,-1}[\tilde{\mu}, m_2] + \frac{1}{2} \overline{C_{\phi2\phi1}} g_2^2 (-3 g_1^2 + g_2^2) \mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2] +$$
$$\frac{1}{2} m_2 \tilde{\mu} g_2^2 (3 g_1^2 (C_{\phi1^2} + C_{\phi2^2}) + 2 C_{\phi2^2} g_2^2) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] + 4 \overline{C_{\phi2\phi1}} g_1^2 g_2^2 \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] +$$
$$\frac{1}{4} g_1^2 g_2^2 (-\tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2}) (7 m_1 + 5 m_2) + 12 \overline{C_{\phi2\phi1}} (m_1 m_2 + \tilde{\mu}^2)) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] +$$
$$2 m_1 m_2 C_{\phi2\phi1} g_1^2 g_2^2 \tilde{\mu}^2 \mathrm{LF}_{2,2,1,0}[m_2, \tilde{\mu}, m_1] - 2 \overline{C_{\phi2\phi1}} g_1^2 g_2^2 \mathrm{LF}_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] +$$
$$g_1^2 g_2^2 (\tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2}) (m_1 + m_2) - 2 \overline{C_{\phi2\phi1}} (m_1 m_2 + \tilde{\mu}^2)) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] -$$
$$2 m_1 m_2 C_{\phi2\phi1} g_1^2 g_2^2 \tilde{\mu}^2 \mathrm{LF}_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] -$$
$$\frac{1}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{l}}^p, m_{\tilde{l}}^s, m_{\tilde{e}}^t] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \overline{y_u}^{st} y_u^{pt} (\overline{a_d}^{pr} a_d^{sr} + 2 \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r] -$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^t] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} (\overline{a_u}^{pr} a_u^{sr} (\overline{y_d}^{st} y_d^{pt} - 2 \overline{y_u}^{st} y_u^{pt}) - \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} (\overline{a_u}^{pr} a_u^{sr} (-\overline{y_d}^{st} y_d^{pt} + 2 \overline{y_u}^{st} y_u^{pt}) + \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r] -$$
$$3 \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \mathrm{LF}_{1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t] -$$
$$\frac{3}{4} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} (a_u^{pt} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} (a_u^{pt} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t] -$$
$$\frac{3}{8} \overline{C_{\phi2\phi1}} (\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} - 3 \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s] +$$
$$\frac{3}{8} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} (a_u^{pt} (C_{\phi1^2} + 3 C_{\phi2^2}) - \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{q}}^s] -$$
$$3 \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \mathrm{LF}_{1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{u}}^r] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{pr} \overline{y_u}^{st} y_u^{pt} a_d^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{d}}^r] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{sr} (\overline{y_d}^{st} y_d^{pt} - 2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r] -$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{sr} (\overline{y_d}^{st} y_d^{pt} - 2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r] -$$
$$\frac{3}{4} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} (3 C_{\phi1^2} + C_{\phi2^2}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^t] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} (3 C_{\phi1^2} + C_{\phi2^2}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^t] -$$
$$\frac{3}{8} \overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{sr} (\overline{y_d}^{st} y_d^{pt} - 4 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p] +$$
$$\frac{3}{8} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} (5 C_{\phi1^2} - C_{\phi2^2}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{q}}^p] -$$
$$3 \overline{y_u}^{st} y_u^{sr} a_u^{pt} (\overline{C_{\phi2\phi1}} \overline{a_u}^{pr} + C_{\phi2^2} \tilde{\mu} \overline{y_u}^{pr}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t] +$$
$$3 \overline{y_u}^{st} y_u^{sr} a_u^{pt} (\overline{C_{\phi2\phi1}} \overline{a_u}^{pr} + C_{\phi2^2} \tilde{\mu} \overline{y_u}^{pr}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t] -$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{q}}^p] +$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{q}}^p] -$$
$$3 \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t] +$$
$$3 \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t] +$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \overline{y_u}^{st} y_u^{sr} \overline{a_u}^{pr} a_u^{pt} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p] +$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^s] -$$
$$\frac{3}{2} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} (a_u^{pt} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt}) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s] +$$
$$\frac{3}{2} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} (a_u^{pt} (C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt}) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s] -$$
$$3 C_{\phi1^2} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r] +$$
$$3 C_{\phi1^2} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r] +$$
$$\frac{3}{2} \tilde{\mu} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{sr} a_u^{pt} (C_{\phi1^2} + C_{\phi2^2}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^t, m_{\tilde{u}}^r, m_{\tilde{q}}^p] -$$
$$\frac{1}{4} g_1^2 g_2^2 (7 \overline{C_{\phi2\phi1}} + 4 \tilde{\mu} (m_1 + m_2)) \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] +$$
$$\frac{1}{2} m_1 g_1^2 g_2^2 (-m_2 \overline{C_{\phi2\phi1}} + \tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2})) \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_1, m_2] +$$
$$3 \overline{C_{\phi2\phi1}} g_1^2 g_2^2 \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + \frac{1}{8} g_1^2 g_2^2$$
$$(-\tilde{\mu} (C_{\phi2^2} (9 m_1 + 5 m_2) + C_{\phi1^2} (31 m_1 + 19 m_2)) + 16 \overline{C_{\phi2\phi1}} (m_1 m_2 + \tilde{\mu}^2)) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] +$$
$$m_1 m_2 C_{\phi2\phi1} g_1^2 g_2^2 \tilde{\mu}^2 \mathrm{LF}_{3,1,1,0}[\tilde{\mu}, m_1, m_2] - \frac{3}{2} \overline{C_{\phi2\phi1}} g_1^2 g_2^2 \mathrm{LF}_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] +$$
$$\frac{3}{4} g_1^2 g_2^2 (\tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2}) (m_1 + m_2) - 2 \overline{C_{\phi2\phi1}} (m_1 m_2 + \tilde{\mu}^2)) \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] -$$
$$\frac{3}{2} m_1 m_2 C_{\phi2\phi1} g_1^2 g_2^2 \tilde{\mu}^2 \mathrm{LF}_{4,1,1,-1}[\tilde{\mu}, m_1, m_2] - 2 \overline{C_{\phi2\phi1}} g_1^2 g_2^2 \mathrm{LF}_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] +$$
$$g_1^2 g_2^2 (\tilde{\mu} (3 C_{\phi1^2} + C_{\phi2^2}) (m_1 + m_2) - 2 \overline{C_{\phi2\phi1}} (m_1 m_2 + \tilde{\mu}^2)) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] -$$
$$2 m_1 m_2 C_{\phi2\phi1} g_1^2 g_2^2 \tilde{\mu}^2 \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] +$$
$$\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{st} (-\overline{C_{\phi2\phi1}} \overline{a_d}^{pr} (y_d^{pt} a_d^{sr} + y_d^{sr} a_d^{pt}) + \tilde{\mu} \overline{y_d}^{pr} y_d^{sr} (a_d^{pt} (-C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pt}))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{q}}^s] - \frac{3}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^4 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr}$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^r, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{q}}^p] - 3 \tilde{\mu}^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s] +$$
$$\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{st} (\overline{C_{\phi2\phi1}} \overline{a_d}^{pr} (y_d^{pt} a_d^{sr} + y_d^{sr} a_d^{pt}) - \overline{y_d}^{pr} a_d^{pt} (C_{\phi2\phi1} a_d^{sr} + \tilde{\mu} y_d^{sr} (-C_{\phi1^2} + C_{\phi2^2})))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^t, m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{q}}^s] +$$
$$\frac{3}{4} C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} a_d^{pt} a_d^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^t, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{q}}^p] +$$
$$\frac{1}{4} \tilde{\mu}^2 \overline{y_e}^{st} (-\overline{C_{\phi2\phi1}} \overline{a_e}^{pr} (y_e^{pt} a_e^{sr} + y_e^{sr} a_e^{pt}) + \tilde{\mu} \overline{y_e}^{pr} y_e^{sr} (a_e^{pt} (-C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pt}))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{e}}^t, m_{\tilde{l}}^s] - \frac{1}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^4 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr}$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^r, m_{\tilde{l}}^s, m_{\tilde{e}}^t, m_{\tilde{l}}^p] - \tilde{\mu}^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p, m_{\tilde{l}}^s] +$$
$$\frac{1}{4} \tilde{\mu}^2 \overline{y_e}^{st} (\overline{C_{\phi2\phi1}} \overline{a_e}^{pr} (y_e^{pt} a_e^{sr} + y_e^{sr} a_e^{pt}) - \overline{y_e}^{pr} a_e^{pt} (C_{\phi2\phi1} a_e^{sr} + \tilde{\mu} y_e^{sr} (-C_{\phi1^2} + C_{\phi2^2})))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^t, m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{l}}^s] +$$
$$\frac{1}{4} C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} a_e^{pt} a_e^{sr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^t, m_{\tilde{l}}^s, m_{\tilde{e}}^r, m_{\tilde{l}}^p] -$$
$$\frac{1}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^s] +$$
$$\frac{1}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} \overline{a_e}^{pr} a_e^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^r, m_{\tilde{e}}^t, m_{\tilde{l}}^s] -$$
$$\frac{1}{2} \tilde{\mu}^2 \overline{y_e}^{st} (\overline{C_{\phi2\phi1}} y_e^{sr} \overline{a_e}^{pr} a_e^{pt} + \overline{y_e}^{pr} (\overline{C_{\phi2\phi1}} \tilde{\mu}^2 y_e^{pt} y_e^{sr} + a_e^{pt} (C_{\phi2\phi1} a_e^{sr} + \tilde{\mu} y_e^{sr} (C_{\phi1^2} + C_{\phi2^2}))))$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^p, m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^s] +$$
$$\frac{1}{2} \tilde{\mu}^2 \overline{y_e}^{st} (\overline{C_{\phi2\phi1}} y_e^{sr} \overline{a_e}^{pr} a_e^{pt} + \overline{y_e}^{pr} (\overline{C_{\phi2\phi1}} \tilde{\mu}^2 y_e^{pt} y_e^{sr} + a_e^{pt} (C_{\phi2\phi1} a_e^{sr} + \tilde{\mu} y_e^{sr} (C_{\phi1^2} + C_{\phi2^2}))))$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^s] - C_{\phi1^2} \tilde{\mu}^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^s, m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p] +$$
$$C_{\phi1^2} \tilde{\mu}^3 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{sr} a_e^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^s, m_{\tilde{e}}^t, m_{\tilde{e}}^r, m_{\tilde{l}}^p] -$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s] +$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} \overline{a_d}^{pr} a_d^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{d}}^t, m_{\tilde{q}}^s] -$$
$$\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{st} (\overline{C_{\phi2\phi1}} y_d^{sr} \overline{a_d}^{pr} a_d^{pt} + \overline{y_d}^{pr} (\overline{C_{\phi2\phi1}} \tilde{\mu}^2 y_d^{pt} y_d^{sr} + a_d^{pt} (C_{\phi2\phi1} a_d^{sr} + \tilde{\mu} y_d^{sr} (C_{\phi1^2} + C_{\phi2^2}))))$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^s] +$$
$$\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{st} (\overline{C_{\phi2\phi1}} y_d^{sr} \overline{a_d}^{pr} a_d^{pt} + \overline{y_d}^{pr} (\overline{C_{\phi2\phi1}} \tilde{\mu}^2 y_d^{pt} y_d^{sr} + a_d^{pt} (C_{\phi2\phi1} a_d^{sr} + \tilde{\mu} y_d^{sr} (C_{\phi1^2} + C_{\phi2^2}))))$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^s] +$$
$$\frac{3}{4} (\tilde{\mu}^3 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} (a_u^{pt} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt}) + \overline{C_{\phi2\phi1}} \overline{a_d}^{pr} a_u^{pt} (a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^r, m_{\tilde{u}}^t] + \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} \overline{a_u}^{pr}$$
$$(2 \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pt} a_u^{sr} + a_d^{pt} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr})) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{u}}^r] -$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{u}}^t] -$$
$$3 \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr} a_u^{pt} a_u^{sr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{u}}^r] - \frac{3}{4} a_u^{pt}$$
$$(2 C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} a_u^{sr} + \overline{a_u}^{pr} (2 \overline{C_{\phi2\phi1}} \overline{a_u}^{st} a_u^{sr} + \tilde{\mu} \overline{y_u}^{st} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr})))$$
$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{u}}^r] + \frac{3}{4} a_u^{pt}$$
$$(2 C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} a_u^{sr} + \overline{a_u}^{pr} (2 \overline{C_{\phi2\phi1}} \overline{a_u}^{st} a_u^{sr} + \tilde{\mu} \overline{y_u}^{st} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr})))$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{u}}^r] + \frac{3}{8} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr}$$
$$(\overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt} a_u^{sr} + a_u^{pt} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + 3 \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr})) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{q}}^s, m_{\tilde{u}}^t] +$$
$$\frac{3}{8} (4 C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} a_u^{pt} a_u^{sr} + \overline{a_u}^{pr} (4 \overline{C_{\phi2\phi1}} \overline{a_u}^{st} a_u^{pt} a_u^{sr} +$$
$$\tilde{\mu} \overline{y_u}^{st} (-\overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt} a_u^{sr} + a_u^{pt} (a_u^{sr} (C_{\phi1^2} + 3 C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr}))))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{q}}^s, m_{\tilde{u}}^r] - 3 C_{\phi1^2} \tilde{\mu}^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p] +$$
$$3 C_{\phi1^2} \tilde{\mu}^3 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{sr} a_d^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{d}}^t, m_{\tilde{d}}^r, m_{\tilde{q}}^p] +$$
$$\frac{3}{2} C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_u}^{st} a_d^{sr} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{d}}^r, m_{\tilde{u}}^t] -$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r, m_{\tilde{u}}^t] -$$
$$\frac{3}{4} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr} a_u^{pt} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{u}}^r] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr} a_u^{pt} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^t, m_{\tilde{u}}^r] +$$
$$\frac{3}{8} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr} (\overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt} a_u^{sr} + a_u^{pt} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) - \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr}))$$
$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{u}}^t] + \frac{3}{8} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr}$$
$$(3 \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{pt} a_u^{sr} + a_u^{pt} (a_u^{sr} (5 C_{\phi1^2} - C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr})) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^s, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{u}}^r] -$$
$$\frac{3}{2} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr} a_u^{pt} (2 C_{\phi1^2} a_u^{sr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr}) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t] +$$
$$\frac{3}{2} \tilde{\mu} \overline{y_u}^{st} \overline{a_u}^{pr} a_u^{pt} (2 C_{\phi1^2} a_u^{sr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr}) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^t] +$$
$$\frac{3}{4} \overline{C_{\phi2\phi1}} \overline{a_u}^{pr} a_u^{sr} (\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^r, m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s] - \frac{3}{2} a_u^{pt} a_u^{sr}$$
$$(C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} + \overline{a_u}^{pr} (\overline{C_{\phi2\phi1}} \overline{a_u}^{st} + \tilde{\mu} \overline{y_u}^{st} (C_{\phi1^2} + C_{\phi2^2}))) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r] +$$
$$\frac{3}{2} a_u^{pt} a_u^{sr} (C_{\phi2\phi1} \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} + \overline{a_u}^{pr} (\overline{C_{\phi2\phi1}} \overline{a_u}^{st} + \tilde{\mu} \overline{y_u}^{st} (C_{\phi1^2} + C_{\phi2^2})))$$
$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^t, m_{\tilde{q}}^p, m_{\tilde{q}}^s, m_{\tilde{u}}^r] - \frac{3}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^t, m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r] +$$
$$\frac{3}{2} \overline{C_{\phi2\phi1}} \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^t, m_{\tilde{q}}^s, m_{\tilde{q}}^p, m_{\tilde{u}}^r] + \frac{3}{4} \tilde{\mu} \overline{y_u}^{st} a_u^{pt}$$
$$(C_{\phi2\phi1} \tilde{\mu} \overline{y_u}^{pr} a_u^{sr} + \overline{a_u}^{pr} (a_u^{sr} (3 C_{\phi1^2} + C_{\phi2^2}) + \overline{C_{\phi2\phi1}} \tilde{\mu} y_u^{sr})) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^t, m_{\tilde{u}}^r, m_{\tilde{q}}^p, m_{\tilde{q}}^s])$$